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RareJob Technologies Uses Amazon SageMaker for English Speaking Tests to Speed Up ML Model Development by More Than 10x

Learn how RareJob Technologies in education made ML model development more efficient using Amazon SageMaker.

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25%

reduction in ML training time



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[RareJob Technologies](#) is the technical department of RareJob Group, which provides the online English conversation service RareJob English Conversation, and it is responsible for product planning and development. The company adopted Amazon Web Services (AWS) to overcome development bottlenecks. In June 2020, it released the English speaking-ability measurement system PROGOS, which achieves automatic scoring using artificial intelligence (AI). Using [Amazon SageMaker](#), the company improved development efficiency by more than 10 times.



Opportunity | Releasing an English Speaking Test That Achieves Automatic Scoring Using AI

The RareJob Group is developing human resource education services for English, such as Smart Method, which is English conversation coaching for corporations, and RareJob English Conversation, which is an English conversation service that has 1 million members, 60 million lessons, and 6,000 instructors.

In June 2020, the company developed the English speaking-ability measurement system PROGOS, which achieves automatic scoring using AI, and began providing the RareJob Speaking Test Powered by PROGOS (speaking test) service for individuals and corporations using PROGOS.

The speaking test measures the examinee's ability to speak English. Examinees take the test online, and it takes about 20 minutes to answer the questions. Examinees are notified of the results in as little as 2–3 minutes, and they know their English proficiency based on international assessment standards. "The speaking test is a service released so that test-takers can grasp their own abilities and use them as a reference for their study plans," says Kentaro Haneda, executive officer and chief technology officer at RareJob Technologies. "Scoring with AI provides a solution for evaluation blurring due to an individual rater's experiences and senses, helping objective results to be announced quickly."

Solution | Using Amazon SageMaker for ML Model Development

The RareJob Group migrated its RareJob English Conversation service from another cloud provider to AWS in 2016. Since then, most of its services have operated on AWS.

“We decided to migrate to AWS because we judged that the bottleneck of our infrastructure would be a hindrance to our business,” says Haneda. “Since most of the lecturers in RareJob English Conversation are in the Philippines, we also evaluated AWS’s region in Singapore as well as its excellent support, technological development, and amount of technical information.”

PROGOS, which began development in 2019, also uses AWS. PROGOS consists of two services: the application side, which asks examinees questions and accepts answers, and the scoring side, which stores the voice data from examinees and scores it with AI. Both services use containers and serverless, managed services such as [Amazon Elastic Container Service](#) (Amazon ECS), [AWS Fargate](#), [Amazon DynamoDB](#) and [AWS Lambda](#) for the runtime environment. To loosely-coupling services on the application side and the scoring side, PROGOS uses [Amazon Simple Queue Service](#) (Amazon SQS).

“Since PROGOS is used for various services, we designed an architecture with scalability in mind,” says Haneda. “Also, we use Amazon SQS to achieve preprocessing launched at the same time that the user starts taking the test. Amazon SQS then quickly transitions to scoring after confirming the completion of the test.”

RareJob Technologies uses Amazon SageMaker to develop ML models for AI scoring, and [AWS Glue](#) and [Amazon Athena](#) to acquire data from examinees for model development. “At first, we were developing ML models on a local PC, but only one model could be launched at a time per developer, which created a bottleneck in development,” says Takuya Yamashiro, ML engineer in the EdTechLab at RareJob Technologies. Yamashiro and his team once tried Amazon EC2 and Amazon ECS, and they needed larger instances with a lower cost. “So, we focused on model training jobs using Amazon SageMaker. We decided to adopt it after seeing that instances with more GPUs and larger memory could be selected relatively freely and that ML could be trained at a lower cost by using things like managed spot training.”

The adoption of Amazon SageMaker significantly improved the efficiency of developing ML models, making it possible to reduce learning time by up to a quarter compared to model development using a local PC, “It is

The RareJob Group is also accelerating its use of AWS in terms of business. Employees transitioned to remote work due to the COVID-19 pandemic, and when the capacity of existing VPNs was insufficient, the company adopted [AWS Client VPN](#) in a short period of time to create a remote work environment for approximately 150 employees. The company also began using [Amazon WorkSpaces](#) for department personnel that handle sensitive management information and personal information. For the routine work of retrieving necessary data from cloud services and uploading files to document management tools, the company developed a system that automatically acquires data using AWS Lambda, reducing the workload by approximately 100 hours per month. “The mission of RareJob Technologies is to increase the value of the RareJob Group, and AWS has established a style that leads to solutions for issues while actively uncovering potential needs,” says Katsuaki Hirakawa, IT solution manager at RareJob Technologies.

Architecture



Outcome | Achieving 3x Growth in Sales Using AWS

It has been about 7 years since the RareJobs Group began using AWS in 2016. Now, AWS is an essential foundation for business.

“In 7 years, the RareJob Group’s sales have almost tripled. In response, internal data use and systems integration have also accelerated, and the size of the engineering department has increased,” says Haneda. “Thanks to AWS, we were able to support a three-fold growth in sales.”

The number of examinees taking the speaking test using PROGOS is also rising steadily, and the number of examinees who choose automatic scoring using AI rather than manual scoring by graders is increasing.

“In recent years, usage by major client companies has been expanding, such as being used to determine the English proficiency of employees and for training new graduates,” says Haneda.

PROGOS is continually evolving while receiving feedback from examinees. Going forward, RareJob Technologies plans to improve the service to be easier to use while introducing a fraud detection system to prevent proxy test-taking. The company is also working on cost reduction using [Amazon SageMaker Serverless Inference](#), which charges only a time inferences run..

The RareJob Group as a whole will promote the efficiency of back-office operations and implement measures to increase the value of products.

“We will actively provide new services while using customer behavior data and lesson data that we have accumulated so far in order to become an AI assessment company,” says Haneda.

About RareJob Technologies

In April 2022, the technical division of the RareJob Group was spun off and established as a subsidiary called RareJob Technologies. The company aims to create an environment where everyone can grow by evolving, updating, and improving the world’s learning through data and technology. Its organization is composed of highly specialized creative groups such as engineers, designers, and planners, and it is responsible for product planning, development, and operation of the RareJob Group.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)



AWS Glue

AWS Glue is a serverless data integration service that makes data preparation simpler, faster, and cheaper.

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Amazon Athena

Amazon Athena is a serverless, interactive analytics service built on open-source frameworks, supporting open-table and file formats.

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AWS Fargate

AWS Fargate is compatible with both Amazon Elastic Container Service (Amazon ECS) and Amazon Elastic Kubernetes Service (Amazon EKS).

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